

MATH 4162/Math 6201

Numerical methods for differential equations

Fall 2025

Case Study #1

Show all necessary steps of working in the given space

Name: _____

ID: _____

Upload the assignment solution no later than

23:59 on Oct 10, 2025

Full marks 50

Many scientific problems in science and engineering deals with partial differential equations, where the solution may exhibit chaotic behaviour. This case study explores **explicit time-stepping schemes** on the **Lorenz '63 system** and asks you to **build** and **validate a reusable numerical solver**. The main objective is to develop a numerical model and use it for a short case study.

- **Develop a numerical model of a chaotic system and verify convergence.**
- **Explore some properties of Lorenz system.**
- **Demonstrate the outcome of scientific inquiries about numerical solution of differential equations.**

Consider the Lorenz system

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z.$$

The parameters are $(\sigma, \rho, \beta) = (10, 28, 8/3)$

Submit your findings through Gradescope and upload a copy of your code to Bright Space Case Study folder.

1. If you have uploaded the code to Brightspace, insert a placeholder page to Gradescope indicating this upload. You may insert a screenshot of a representative portion of your code, which is not needed. However, upload a full copy to the assignment folder. The uploaded code should be functional i.e. would run as is, which does not need to reproduce all figures reported below.
2. Implement Forward Euler method and Runge Kutta (RK4) method for solving Lorenz system. This will be a reusable code that you should be able to apply the same code, without needing modification, to solve any other differential equations whenever the current methods are to be applied.

Compare Lorenz trajectory as a 3D plot between the two methods for $t \in [0, 100]$ and any initial condition.

Briefly discuss what you observed regarding convergence as $\Delta t \rightarrow 0$. Using lead error of any of the method, explain the convergence behaviour that you observed.

3. Compare the solutions obtained by two methods. Plot t vs $x(t)$, t vs $y(t)$, and t vs $z(t)$. For each plot, you will show and compare two curves due to two methods. The time step and initial condition are your choice, but the submission should state the value of Δt .

4. **(Math 4162 only)** Consider initial condition to be a random vector $[x(0), y(0), z(0)]^T$ in $[-30, 30]$. Take 100 random initial conditions from $[-30, 30]$. For each initial condition, simulate a Lorenz trajectory using Euler method. Display 100 trajectories in a single 3D plot. For each trajectory, use marker to indicate initial condition, and solid line to show the trajectory. Repeat this for RK4 method. You will submit two plots.

Briefly mention your observation about the invariant structure of the solution and its sensitivity to numerical method and time step.

5. **(Math 6201 only)** The stability and well-posedness of any linear PDE requires that

$$||\hat{u}(k, t)|| \leq ||\hat{u}(k, 0)||\kappa e^{\alpha t}$$

for some κ and α . (See lecture 2.) An analogous diagnostic for the nonlinear Lorenz system is to examine the separation between nearby trajectories.

Let $\mathbf{p}(t)$ and $\mathbf{q}(t)$ denote two trajectories for two initial conditions: $\mathbf{p}(0)$ and $\mathbf{q}(0)$. Thus, we expect that

$$\delta(t) \sim ||\delta(0)||e^{\alpha t},$$

where $\delta(t) = ||\mathbf{p}(t) - \mathbf{q}(t)||$.

Simulate this behaviour by considering $\mathbf{p}(0)$ and $\mathbf{q}(0)$ to be very close, and submit plot of t vs $\ln \delta(t)$ and comparison of the results between Euler and RK4 methods, using the same Δt and total time of 100. Also, estimate the value of α .